

YI GU

(713)-380-7338 ◇ Yi.Gu@u.northwestern.edu ◇ wildskywalker.github.io

EDUCATION

Northwestern University

Ph.D in Mathematics

Advisor: Antonio Auffinger

Thesis Title: Generalized t-SNE and Beyond: Probabilistic Methods for Dimensionality Reduction, Combinatorial Optimization, and Machine Learning

Rice University

B.S. in Mathematics, President Honor Roll

EXPERIENCE

Google

Sept 2024 - Current

Research Data Scientist - AI System Knowledge and Information Engine (ASKIE)

February 2026–present

New York City, NY

Joined shortly after Gemini 3.1 Pro's launch; tenure spans the introduction of Gemini 4 Argon.

- Lead the Work Done by AI team, designing and owning **Google-wide metrics** of AI contribution, primarily focused on Antigravity coding agents powered by multiple models, including Gemini 4 Argon.
- Own the development of **grading algorithms** for Google Antigravity coding-agent trajectories, measuring effort-equivalent AI contribution, correctness, complexity, and impact; completed trajectory analysis intended for use by Google DeepMind and other internal teams.
- Developed a modular statistical-reporting and visualization skill in Gemini's internal codebase, used by **500+ Google employees** and integrated into team data-science and organizational insight agents; supports standalone use and composition within larger skill suites, and serves as an internal example of effective skill design.
- Conduct cross-product studies of AI adoption, employee behavior, and productivity across Google, applying double machine learning and continuous-treatment difference-in-differences to inform development priorities for next-generation coding agents. Analyze hundreds of billions of first-party coding-agent and LLM usage records.
- Co-organize Google's internal AI study group and invited-speaker series, and deliver workshops on causal inference and machine learning to data-science teams.

Research Data Scientist - Google Hardware (Pixel)

September 2024–January 2026

Chicago, IL

- Built an LLM sentiment-analysis pipeline using LLM-as-judge, SFT, and RAG; achieved 99.5% accuracy and informed hardware KPI targets and RMA analysis.
- Developed a Pixel diagnostic pipeline using LoRA and RLHF; improved classification accuracy from 40% to over 90% (0.95+ recall) and increased throughput by 300%.

Google

June 2022 – Sept 2022 & June 2023 – Sept 2023

Data Scientist Intern (Ph.D)

Mountain View, CA

- Built a failure-prediction pipeline achieving over 90% recall and reducing system failures by over 50%; implemented homomorphic encryption algorithms with Google Ads for privacy-preserving advertising.

PUBLICATIONS

(Preprint) Lingyou Pang, Jianyu Lin, **Yi Gu**^{*}, *et al.* *LLM-Judge Autopilot: Analytically Grounded Reliability and Uncertainty for LLM-as-a-Judge*.

(Preprint) **Yi Gu**, Lingyou Pang, Xiangkun Ye, *et al.* *SIGMA: Scalable Spectral Insights for LLM Model Collapse*. arXiv:2601.03385

(Preprint) **Yi Gu**. *Equilibrium Distribution for t-Distributed Stochastic Neighbor Embedding with Generalized Kernels*. arXiv:2505.24311

Nan Jiang, **Yi Gu**^{*}, *et al.* *Learning Combinatorial Structures via Markov Random Fields with Sampling through Lovász Local Lemma*. **AAAI 2023**

Zichao Wang, **Yi Gu** *et al.* *Iterative Imputating Variational Auto-Encoders for Partially Observed Data*. **EDM 2020**

(Book) **Yi Gu**. *Introduction to Random Graphs and Models*. Research Survey in Random Graph, 2019. (PDF link available)

ACADEMIC SERVICE & MENTORSHIP

ICML 2026	Gold Reviewer (top 25%)	
Peer review	NeurIPS 2026; AAAI 2027; AISTATS 2027	
Google	<i>Intern Mentor</i>	2025
	Guided a Data Scientist intern to a full-time return offer.	
Northwestern University	<i>Instructor, Random Matrix Theory</i>	2019
Rice University	<i>Lead Organizer and Speaker, Spectral Theory Seminar</i>	2016
	Prepared and delivered 9 of 12 graduate lectures on OPUC.	

TECHNICAL SKILLS

Programming	Python, SQL, MATLAB, Mathematica
ML/DL Frameworks	PyTorch, TensorFlow, JAX, Scikit-learn, Pandas, NumPy
Platforms/Tools	Google Cloud (BigQuery, Vertex AI), Apache Beam (Google Flume), Kubernetes (Google Borg), Docker
Expertise	Large Language Models (LLMs), Anomaly Detection, Dimensionality Reduction and Data Visualization, Statistical Modeling, Time Series Analysis, Stochastic Analysis, Pattern Recognition, Combinatorial Optimization, Supervised Fine-Tuning (SFT), Retrieval-Augmented Generation (RAG)

^{*}Equal contribution, co-first author